Project Report: Augmentation and Participant-Normalized Wellbeing Machine Learning

1 Introduction

This report explains the main experiments used in our wellbeing machine learning project. We tested both supervised and unsupervised learning approaches. We also studied the effect of data augmentation and participant-normalized features.

The main goal was to classify or cluster wellbeing states such as normal, sleepy, stress, and abnormal conditions using sensor-window features.

2 Supervised Learning

For supervised learning, we used a scientifically correct evaluation method called Leave-One-Subject-Out validation.

```
Leave-One-Subject-Out validation using video_source
```

This means that each fold holds out one participant for testing:

```
Train: 3 participants
Test: 1 unseen participant
```

This evaluation checks whether the model can generalize to a participant that it has never seen before.

2.1 Supervised Data Augmentation

We added Gaussian jitter augmentation in a fold-aware and participant-aware way.

The most important rule was:

> Augmentation was applied only inside the training fold. The held-out participant was never augmented and was never used to compute augmentation statistics.

For supervised tasks, augmentation was done separately by participant and class.

```
participant + class
```

For example:

```
Alessia normal -> augmented only from Alessia normal
Alessia sleepy -> augmented only from Alessia sleepy
Mo is not used if Mo is held out
```

This prevents data leakage from the test participant into the training data.

2.2 Supervised Results

Task	Baseline Macro F1	Best Augmented Macro F1	Change
Binary normal/abnormal	0.952	0.957	+0.005
Subtype sleepy/stress	1.000	1.000	0.000
Hierarchical	0.957	0.960	+0.003
Direct 3-class	0.951	0.957	+0.006

Table 1: Supervised learning results with and without augmentation.

2.3 Interpretation

Supervised augmentation improved the results only slightly. This is because the supervised baseline was already very strong. The model had already learned the class labels well, so small Gaussian jitter did not change the decision boundary much.

Therefore, augmentation was useful, but it was not the main reason for strong supervised performance.

3 First Unsupervised Approach: Raw Clustering

For unsupervised learning, labels were not used during model training. We tested different clustering methods:

```
K-Means
Gaussian Mixture
Agglomerative Clustering
```

The labels were used only after clustering to evaluate how well the clusters matched the real wellbeing states.

3.1 Raw Unsupervised Results

Task	Best Raw Method	Macro F1	Balanced Accuracy
Binary clustering	Gaussian Mixture	0.810	0.860
3-class clustering	Gaussian Mixture	0.815	0.822

Table 2: Raw unsupervised clustering results.

3.2 Interpretation

The raw clustering results were not perfect. The best method was Gaussian Mixture, but many abnormal, sleepy, or stress samples were still grouped incorrectly.

This probably happened because the clustering model was partly learning participant identity instead of wellbeing state. In other words, raw sensor values can be different from person to person, even when they are in the same wellbeing state.

4 Unsupervised Augmentation

For unsupervised clustering, augmentation was label-free.

The important rule was:

> We did not augment separately by true label. We augmented only by participant, using `video_source`.

So the unsupervised augmentation process was:

```
Alessia rows -> augmented from Alessia rows
Mo rows -> augmented from Mo rows
No mixing between participants
No use of true_label during augmentation
```

This keeps the unsupervised setting correct because true labels are not used to generate augmented samples.

4.1 Unsupervised Augmentation Results

Task	Raw Baseline Macro F1	Raw + Augmentation Macro F1	Change
Binary clustering	0.810	0.871	+0.061
3-class clustering	0.815	0.850	+0.035
LOSO anomaly detection	0.806	0.765	-0.041

Table 3: Unsupervised clustering results with augmentation.

4.2 Interpretation

Augmentation helped normal clustering, especially for the binary clustering task. However, it did not help LOSO anomaly detection. In fact, for anomaly detection, augmentation reduced performance.

Because of this, augmentation should not be kept as the main result for anomaly detection.

5 Main Improvement: Participant-Normalized Features

The biggest improvement came from changing the feature representation.

Instead of clustering raw feature values, we transformed each feature relative to each participant's own normal baseline.

For each participant, we calculated the mean and standard deviation of their normal samples:

```
participant_normal_mean
participant_normal_std
```

Then each feature was converted into a participant-level z-score:

$$z = \frac{value - participant_normal_mean}{participant_normal_std}$$

This means the model sees:

```
How far is this window from this participant's normal state?
```

instead of only seeing:

```
What is the raw HR, head tilt, or blink value?
```

This is important because each participant can have a different normal baseline.

5.1 Participant-Normalized Results

Task	Raw Baseline Macro F1	Participant Z-Score Macro F1	Change
Binary clustering	0.810	0.971	+0.161
3-class clustering	0.815	0.943	+0.128

Table 4: Improvement from participant-normalized z-score features.

The best binary participant-normalized confusion matrix was:

$$\begin{bmatrix} 218 & 0 \\ 17 & 414 \end{bmatrix}$$

The best 3-class participant-normalized confusion matrix was:

$$\begin{bmatrix} 218 & 0 & 0 \\ 0 & 197 & 16 \\ 16 & 5 & 197 \end{bmatrix}$$

These results show that participant-normalized features greatly improved cluster separation.

6 Important Caveat

It is important to explain that participant-normalized unsupervised clustering is not exactly the same evaluation setting as supervised LOSO classification.

Supervised LOSO asks the question:

> Can the model predict a completely unseen participant without using their data?

Participant-normalized clustering asks a different question:

> If we have a normal calibration baseline for each participant, can we cluster their later states better?

Therefore, if unsupervised binary clustering reaches a Macro F1 score of 0.971, this does not mean that unsupervised learning is always better than supervised learning.

The unsupervised participant-normalized method uses a stronger assumption:

> We know each participant's normal baseline.

This assumption is realistic if the system has a calibration phase, but it is different from strict unseen-subject generalization.

7 How to Explain the Interesting Result

An interesting result is that participant-normalized unsupervised binary clustering achieved a higher Macro F1 score than the supervised binary model.

However, this should be explained carefully:

> Interestingly, participant-normalized unsupervised binary clustering achieved higher Macro F1 than the supervised binary model. However, this does not mean that unsupervised learning is generally better. The unsupervised method used each participant's own normal baseline as a calibration reference, while supervised LOSO tested prediction on an unseen participant without such calibration. Therefore, the two results answer different scientific questions.

8 Final Conclusion

Supervised LOSO classification remains the strongest approach for predicting unseen participants without calibration. Data augmentation produced only small supervised gains because the baseline was already strong.

For unsupervised learning, raw clustering was limited because it was affected by participant-specific differences. Label-free augmentation helped clustering slightly, but it was not the main improvement.

The largest improvement came from participant-normalized z-score features. These features represent each sensor window as a deviation from the participant's own normal baseline. This reduces individual differences and helps the clusters better reflect wellbeing states rather than participant identity.

> Overall, the results suggest that wellbeing states are better represented as deviations from each participant's normal baseline rather than as raw sensor values.

9 validating after removing HR and RMSSD/RMMSD

I also tested whether the supervised LOSO models can work without heart-sensor features.

A separate experiment script was used:

`/Users/mosi_dev/Developer/university/nearable-project/Scripts/ML/supervised_no_hr_rmmsd_experiment.py`

The outputs were saved here:

```
/Users/mosi_dev/Developer/university/nearable-project/results/no_hr_rmmsd
```

This experiment tested how important the heart-related features are for supervised classification. The removed features were:

```
HR
RMMSD
```

Two scenarios were tested.

9.1 Scenario 1: Clean No-Heart-Sensor Model

In this scenario, the model was trained and tested using only posture and blink/eye features. The HR and RMMSD features were completely removed from both training and testing.

This is the scientifically safer setting for testing whether the system can work without heart-sensor data.

9.2 Scenario 2: Masked Test Stress Test

In this scenario, the model was trained using all 18 features. However, during the LOSO test folds, HR and RMMSD were set as missing values. Then, the fold's imputer filled the missing values.

This scenario checks what happens if the heart-sensor features are available during training but missing during testing.

However, this is a deployment mismatch because the training and testing feature conditions are different. Therefore, for a real missing-sensor claim, the clean no-heart-sensor model is the more reliable result.

9.3 Main Results

Task	All Features Macro F1	No HR/RMMSD Macro F1	Change
Binary normal/abnormal	0.952	0.954	+0.002
Subtype sleepy/stress	1.000	0.691	-0.309
Hierarchical	0.957	0.761	-0.196
Direct 3-class	0.951	0.750	-0.201

Table 5: Supervised LOSO results after removing HR and RMSSD/RMMSD features.

9.4 Interpretation

The binary normal-vs-abnormal task still worked well without HR and RMMSD. In fact, the Macro F1 score stayed almost the same:

```
All features: 0.952
No HR/RMMSD: 0.954
```

This suggests that posture and blink/eye features are enough for coarse abnormality detection.

However, the subtype sleepy-vs-stress task dropped strongly:

```
All features: 1.000
No HR/RMMSD: 0.691
```

The hierarchical and direct 3-class models also dropped a lot. This means that HR and RMSSD/RMMSD features are important for distinguishing between abnormal subtypes, especially sleepy and stress.

The masked test version performed slightly better than the clean no-heart-sensor model for subtype, hierarchical, and direct 3-class tasks. However, because it trains with all features and tests with missing heart features, it is less realistic for deployment.

Therefore, the clean model trained and tested without HR/RMMSD should be used as the main result for the missing-heart-sensor analysis.

9.5 Conclusion of This Experiment

When HR and RMSSD/RMMSD were removed, binary normal-vs-abnormal detection remained stable. This suggests that posture and blink features are sufficient for coarse abnormality detection.

However, sleepy-vs-stress and 3-class classification dropped substantially. This indicates that heart-rate and HRV features are important for distinguishing abnormal subtypes.

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